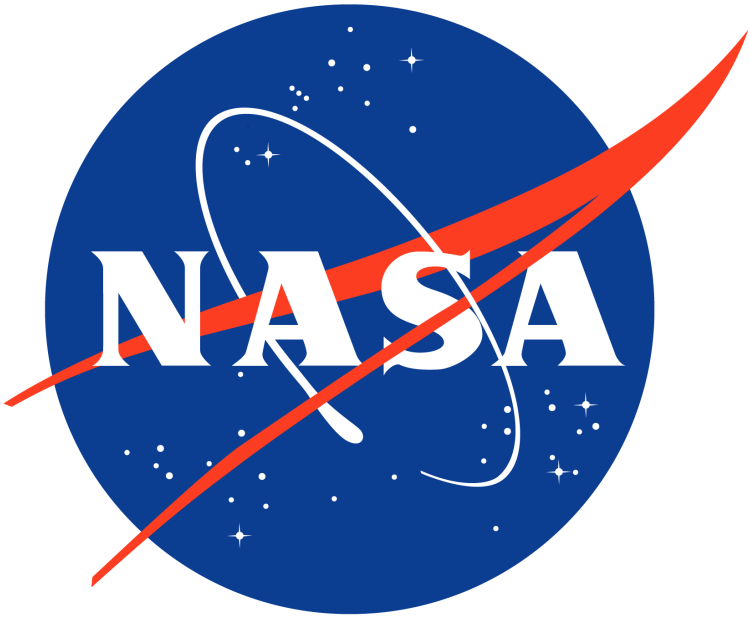


CCS Climate Summer School

2024 Group Projects

Quick Setup Guide



Jet Propulsion Laboratory
California Institute of Technology

Alex Goodman

08/01/24

Important links

- <https://jpl-cmda.org/ccs.html> – Group Projects Page
- <https://hub.jpl-cmda.org> – Jupyter Lab Server

Jupyter Login

- Username: your email address
- Password: nasa-ccs

Sign in

Username:

Password:

Sign in

<https://hub.jpl-cmda.org>

Environment Selection

Server Options



Default

Default Environment for 2024 NASA Summer School Group Projects



RCMES

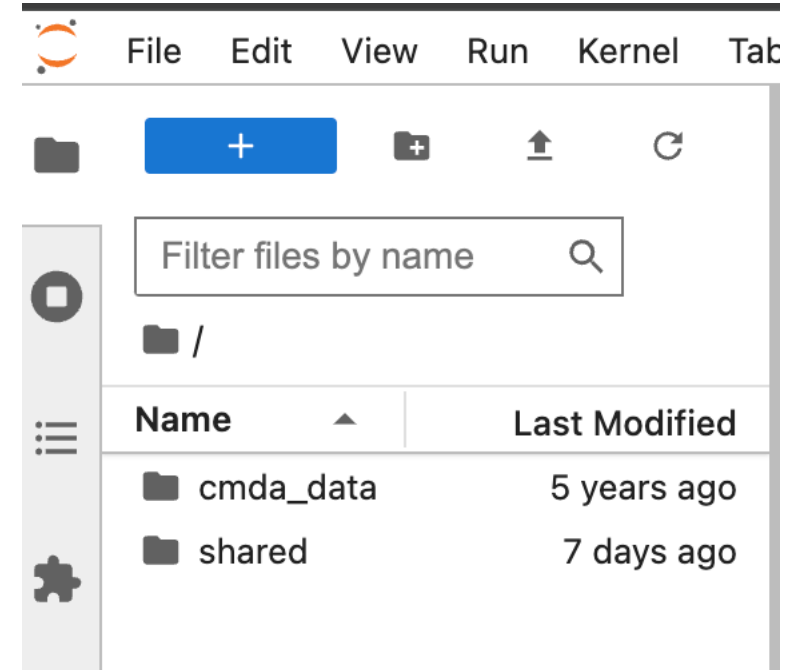
Special Environment for High Resolution Regional Climate Model Project

Start

Select the “Default” environment (unless you select the regional climate modeling topic)

Basic Navigation

- **cmda_data/** - Contains a large variety of model output results (including CMIP5/6), reanalysis, and satellite observations.
 - See: <https://jpl-cmda.org/datasets.html>
 - When using the CMDA interface to search for data, replace data/data_clean with cmda_data/ to get the path to the data on the server.
- **shared/NASA_Summer_School_2024** – Group specific notebooks, scripts, and domain specific datasets



Group Projects Folder

- Under shared/NASA_Summer_School_2024:
 - Select from one of six group projects
- For each project:
 - **materials/** - This includes additional data, notebooks, and scripts needed to complete the group project and is **read-only**.
 - **work/** - This directory is writeable and can be used to place your own notebooks, code, and data. **This will be accessible to all users on the server.**



/ shared / NASA_Summer_School_2024 /	
Name	Last Modified
global_warming	7 days ago
marine_co2	7 days ago
regional_climate	7 days ago
sea_level_rise	7 days ago
swot	7 days ago
tree_mortality	7 days ago

/ ... / NASA_Summer_School_2024 / global_warming /	
Name	Last Modified
materials	5 hours ago
work	7 days ago

Quick Demo

Filter files by name

/ ... / global_warming / materials /

Name	Last Modified
AVHRR_OI-NCEI-L4-GL...	22 hours ago
AVHRR_OI-NCEI-L4-GL...	22 hours ago
NODC	21 hours ago
global_warming.ipynb	5 hours ago

Launcher
global_warming.ipynb

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Download Markdown

Python 3 (ipykernel)

Topic 1: Where is the global warming?

Question 1: Calculate the global net radiative flux imbalance at TOA

Net radiative flux at TOA is calculated from:

$$\Delta F = F_{SW}^{\downarrow} - F_{SW}^{\uparrow} - F_{LW}^{\uparrow}$$

Where $F_{SW}^{\downarrow} - F_{SW}^{\uparrow}$ is the net incoming shortwave radiation and $F_{LW}^{\uparrow}$ is the outgoing longwave radiation at TOA respectively.

First, let's load each of these from the CERES satellite instrument data:

```
[ ]: import xarray as xr
import numpy as np
import matplotlib.pyplot as plt
from cmda_utils import plot_map

rsdt = xr.open_dataset('/home/jovyan/cmda_data/cmip5/nasa/ceres/nasa_cer...
rsut = xr.open_dataset('/home/jovyan/cmda_data/cmip5/nasa/ceres/nasa_cer...
rlut = xr.open_dataset('/home/jovyan/cmda_data/cmip5/nasa/ceres/nasa_cer...
```

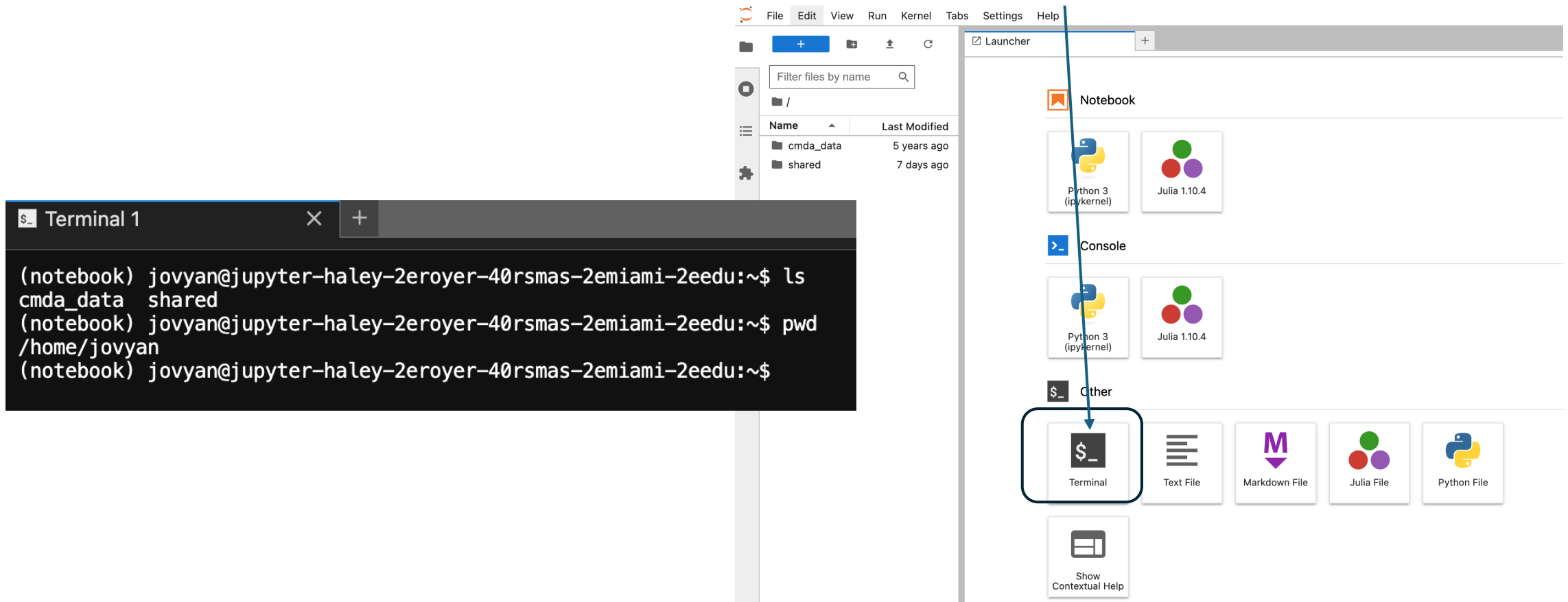
The net radiative flux is therefore:

Other important notes about data sharing

- All data under **cmda_data/** and **shared/** are mounted on separate disks and are shared globally with all users.
- Your home directory (the root of the file explorer, /home/jovyan) has a separate, private storage allocation of 10 GB. You may place your own scripts and notebooks in here for testing purposes, but **PLEASE place any additional data you generate under your group folders only (shared/NASA_Summer_School_2024/<your_topic>/work/)**
- **We HIGHLY recommend adding your own personal work folder to this shared area** (navigate to the directory above, right click the file explorer, select “New Folder”, name it your username / email address. Use this for writing your notebooks / scripts.

Advanced: Command Line

- Select File > New > Terminal (or select “Terminal” in the launcher)



The image shows the JupyterLab interface. On the left, a terminal window titled "Terminal 1" displays the following commands and output:

```
(notebook) jovyan@jupyter-haley-2eroyer-40rsmas-2emiami-2eedu:~$ ls  
cmda_data  shared  
(notebook) jovyan@jupyter-haley-2eroyer-40rsmas-2emiami-2eedu:~$ pwd  
/home/jovyan  
(notebook) jovyan@jupyter-haley-2eroyer-40rsmas-2emiami-2eedu:~$
```

On the right, the "Launcher" sidebar is visible. It contains three sections: "Notebook" with "Python 3 (ipykernel)" and "Julia 1.10.4"; "Console" with the same two options; and "Other" with "Terminal", "Text File", "Markdown File", "Julia File", and "Python File". A blue arrow points from the "Terminal" icon in the "Other" section to the terminal window on the left.

Need additional packages?

- If you are using python, you can make temporary updates locally via running conda or pip (or Pkg if using Julia)
- These changes will not be permanent and will need to be installed again if you restart your server (File > Hub Control Panel > Stop My Server). To request additional python packages with a permanent installation, contact me (but please test your changes locally first!)

Questions

- Alex Goodman (alexander.goodman@jpl.nasa.gov)
- Seungwon Lee (seungwon.lee@jpl.nasa.gov)